

Heat

MCQ

1. The energy possessed by particles due to their motion is called
 - A) Sound energy
 - B) Light energy
 - C) Heat energy
 - D) Chemical energy

Answer: C

2. Heat is transferred because of difference in
 - A) Mass
 - B) Shape
 - C) Temperature
 - D) Size

Answer: C

3. Which type of energy transfer does not need a medium?
 - A) Conduction
 - B) Convection
 - C) Radiation
 - D) Expansion

Answer: C

4. Heat transfer in solids mainly occurs by
 - A) Convection
 - B) Radiation
 - C) Condensation
 - D) Conduction

Answer: D

5. Which of the following is a good conductor of heat?
 - A) Plastic
 - B) Rubber
 - C) Copper

D) Wood

Answer: C

6. The process of heat transfer involving movement of fluid particles is

A) Radiation

B) Conduction

C) Reflection

D) Convection

Answer: D

7. Snow houses are warm inside because snow is a

A) Good conductor

B) Heat producer

C) Poor conductor

D) Reflector

Answer: C

8. Increase in temperature indicates increase in

A) Density

B) Mass

C) Kinetic energy

D) Volume

Answer: C

9. Expansion of a substance occurs when

A) Heat is removed

B) Heat is added

C) Pressure increases

D) Size decreases

Answer: B

10. Which change is a physical change?

A) Burning

B) Rusting

C) Melting

D) Cooking

Answer: C

11. Solid changing directly into gas is called

A) Evaporation

B) Condensation

C) Sublimation

D) Deposition

Answer: C

12. Liquid changing into gas is known as

A) Freezing

B) Melting

C) Vapourisation

D) Deposition

Answer: C

13. Gas changing into liquid is called

A) Condensation

B) Sublimation

C) Melting

D) Freezing

Answer: A

14. Which state of matter has maximum kinetic energy?

A) Solid

B) Liquid

C) Gas

D) Plasma

Answer: C

15. Water exists naturally in how many states on Earth?

A) One

B) Two

C) Three

D) Four

Answer: C

16. The movement of heat from hot to cold region stops when

A) Cold becomes hot

B) Temperature becomes equal

C) Heat disappears

D) Substance expands

Answer: B

17. Which metal is commonly used to make calorimeters?

A) Iron

B) Copper

C) Steel

D) Lead

Answer: B

18. The principle of calorimetry is based on

A) Expansion of solids

B) Radiation law

C) Conservation of energy

D) Laws of motion

Answer: C

19. Heat lost by hot body is equal to

A) Heat destroyed

B) Heat produced

C) Heat gained by cold body

D) Temperature change

Answer: C

20. The SI unit of heat energy is

A) Calorie

B) Kelvin

C) Joule

D) Watt

Answer: C

21. One calorie is equal to

A) 2.186 J

B) 3.186 J

C) 4.186 J

D) 5.186 J

Answer: C

22. Temperature is measured using

A) Calorimeter

B) Thermometer

C) Barometer

D) Ammeter

Answer: B

23. The SI unit of temperature is

A) Celsius

B) Fahrenheit

C) Kelvin

D) Joule

Answer: C

24. Which scale is most commonly used in scientific calculations?

A) Celsius

B) Fahrenheit

C) Kelvin

D) Rankine

Answer: C

25. Heat capacity depends on

A) Colour

B) Shape

C) Mass and material

D) Volume only

Answer: C

26. Specific heat capacity is defined for

A) Any quantity

B) One kilogram only

C) One gram of substance

D) One litre of substance

Answer: C

27. The symbol used for heat capacity is

A) T

B) H

C) C

D) Q

Answer: C

28. Substances with high specific heat

A) Heat up quickly

B) Cool quickly

C) Change temperature slowly

D) Melt easily

Answer: C

29. Which part of thermos flask reduces heat loss by radiation?

A) Vacuum

B) Stopper

C) Silvered walls

D) Plastic cover

Answer: C

30. Vacuum between walls prevents heat transfer by

A) Radiation only

B) Conduction only

C) Convection only

D) Conduction and convection

Answer: D

31. Black surfaces are preferred for cooking vessels because they

- A) Reflect heat
- B) Absorb heat
- C) Prevent conduction
- D) Reduce weight

Answer: B

32. White clothes are worn in summer because they

- A) Absorb heat
- B) Produce heat
- C) Reflect heat
- D) Trap heat

Answer: C

33. Which of the following is an insulator?

- A) Aluminium
- B) Iron
- C) Glass
- D) Copper

Answer: C

34. Freezing is the change from

- A) Gas to solid
- B) Liquid to solid
- C) Solid to liquid
- D) Gas to liquid

Answer: B

35. Deposition is the change from

- A) Liquid to gas
- B) Solid to gas
- C) Gas to solid

D) Liquid to solid

Answer: C

36. Which process causes land and sea breeze?

A) Radiation

B) Expansion

C) Conduction

D) Convection

Answer: D

37. Hot air balloons rise due to

A) Radiation

B) Conduction

C) Convection

D) Reflection

Answer: C

38. Refrigerator works mainly on the principle of

A) Radiation

B) Convection

C) Expansion

D) Reflection

Answer: B

39. Heat from a fire reaches us mainly by

A) Conduction

B) Convection

C) Radiation

D) Evaporation

Answer: C

40. The bottom of cooking vessels is painted black to

A) Look attractive

B) Increase conduction

C) Absorb more heat

D) Reduce weight

Answer: C

41. Thermostat helps to

A) Measure heat

B) Store heat

C) Maintain constant temperature

D) Transfer heat

Answer: C

42. Thermostat works using

A) Chemical reaction

B) Temperature change

C) Light reflection

D) Sound waves

Answer: B

43. Fan regulator uses

A) Radiation

B) Thermostat principle

C) Calorimetry

D) Expansion only

Answer: B

44. Which material is used to prevent heat loss in calorimeter?

A) Copper

B) Aluminium

C) Insulating jacket

D) Steel

Answer: C

45. Stirrer in a calorimeter is used to

A) Heat liquid

B) Measure mass

C) Ensure uniform temperature

D) Prevent radiation

Answer: C

46. Heat transfer through empty space is possible due to

A) Convection

B) Conduction

C) Radiation

D) Expansion

Answer: C

47. Increase in heat generally causes substances to

A) Contract

B) Expand

C) Freeze

D) Condense

Answer: B

48. The temperature of an object depends on

A) Total heat

B) Average kinetic energy

C) Mass only

D) Volume only

Answer: B

49. Which is NOT a method of heat transfer?

A) Conduction

B) Convection

C) Radiation

D) Insulation

Answer: D

50. Thermos flask is mainly used to

A) Measure temperature

B) Produce heat

C) Maintain temperature

D) Increase heat transfer

Answer: C